

Sealed Experiment Record

Sealed 2026-10-07 · Signed by the K-Veritas server
Session 144b2aed · Data hash 7a4a...83e7

Abstract

This record seals 1 run (kv_run.sh) executed on Intel(R) Xeon(R) W-2145 CPU @ 3.70GHz between 2026-10-07 07:37 and 07:43 UTC. Reported results: cifar_eval_acc_pct 93.75, mnist_test_acc_pct 98.31. 1 invocation in total, 0 failed. Produced with K-Veritas [1].

What this record proves

These numbers came from the sealed code at this time, unchanged since. Whether the experiment is sound is for the reader to judge.

Verify: kveritas verify <this file> or kveritas.org/verify

1 Setup

Item	Value
Session	144b2aed-1eae-4b7e-8638-e52ee65e5910
Initialized	2026-10-07T07:37:16Z
Machine	167ec0d6dc9a8170
Operating system	linux / amd64
CPU	Intel(R) Xeon(R) W-2145 CPU @ 3.70GHz, 16 cores
Memory	31.0 GB
GPU	NVIDIA GeForce RTX 5060 Ti, 16311 MiB
Environment, run 1	9e2e5283311d84149f525c2b5c322185daf1aff859a0fca9e9c03597facaf7a6
Disclosure	open
Source code, run 1	7bfc712a0b4c9c58995409d495276d9a52953879cbecb0ad5368e5805d45c4c2

2 Runs

#	Command	Duration	Exit	Anchored
1	bash kv_run.sh	5 minutes 58 seconds	0	at run end

1 invocation in total: 1 kept in this record, 0 failed, 0 interrupted. Every invocation is recorded by the server, including the ones not kept.

3 Results

Run 1: kv_run.sh

Metric	Final value	Step	Source
cifar_train_loss	508.86	950 (20 readings)	explicit
cifar_eval_acc_pct	93.75	1000	explicit
mnist_final_loss	0.08		explicit

Inline claim	Value	Phase	Output line
cifar_eval_acc_pct	93.75		1037
mnist_test_acc_pct	98.31		1113

Seed committed: seed:201, output line 1, 2026-10-07T07:37:17Z.

4 Execution evidence

Coherence

Coherent (0.50): the telemetry channels move together, as one process.

Telemetry

Run	Samples	CPU time	Peak memory	Peak GPU
1	3566	377.0 s	2.29 GB	99%

5 Provenance

Run 1: kv_run.sh

Disclosure open. 1147 tracked files, 4 snapshots. Every snapshot is bound into the signature.

#	Snapshot	Time	Root	Changed
0	run_start	07:37:17	3e0eb97765d9d1b1	1147
3	run_end	07:43:14	3e0eb97765d9d1b1	0

2 snapshots with no file change omitted here; full timeline in Appendix B.

Withheld by .kveritasignore	Size	Hash
docs/abstractions3.py	<1MB	4a8ec2016f88078153f74db1
docs/abstractions4.py	<1MB	85e09401595ed2631fe8e310
spec/render.sh	<1KB	380d043f58396075ed545139
test/__init__.py	<1KB	9de1c3cc51b614f0368b15b9
test/amd/__init__.py	<1KB	248bfdef67451ef05d09c3f9
test/amd/disasm.py	<1MB	1f10083898281e6cac17ce28
test/amd/helpers.py	<1MB	7ebce584d05fcb2085750754
test/amd/hw/__init__.py	<1KB	dcd716649a3f391cb9304b4c
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test/null/test_tensor_uop_mixin.py	<1MB	78e8bce991cb3cd8eed594bf
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test/null/test_transcendental_helpers.py	<1MB	8531550c63b1e2bc8aa2dd27
test/null/test_uir_render.py	<1MB	25ae5bb5d3145ade1f723488
test/null/test_uop_graph.py	<1MB	bcb5460654dc2665f7679032
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test/null/test_uop_symbolic.py	<1MB	ce8021283805a9596db5c78f
test/null/test_uop_vmin_vmax.py	<1MB	05d9d80766c8e17ae58938d5
test/null/test_uops.py	<1MB	099233d13fa2e2d23362f6fd
test/null/test_uops_stats.py	<1MB	6ce47bc1ec1e972dcd6a6a87
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test/null/test_viz.py	<1MB	f93684254d8c77cbc478ba7a
test/null/test_winograd.py	<1MB	1a2c9c9839340c910c1997ea
test/runtime/__init__.py	<1KB	8122bdaa2973b01d99b77bf6
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test/runtime/test_allreduce.py	<1MB	06075867eff15a81253bc9fb
test/runtime/test_arange.py	<1MB	4482ce9b1df5ddc46bdba511
test/runtime/test_asm_gemm.py	<1MB	51e9d3ff115a2e6fdd1a8783
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test/runtime/test_conv.py	<1MB	92e68329e4ea2714e474d006
test/runtime/test_custom_kernel.py	<1MB	3d5627547b3ceb4cb5723d7e
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test/runtime/test_linalg.py	<1MB	0b59a045dabdc5510741cd11
test/runtime/test_linearizer.py	<1MB	c8275075a61c30536bacddba
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test/runtime/test_llm_moe.py	<1MB	f937a49fe870835476760910
test/runtime/test_llm_quantized.py	<1MB	bf2e8942e56cc50bf43cc874
test/runtime/test_llm_server.py	<1MB	ca78700c40b18b9e09e3353b
test/runtime/test_llm_tp.py	<1MB	555ec78d06b7fae2d074973e
test/runtime/test_masked_tensor.py	<1KB	fd9eb5a92ebcf7be0f9231cf
test/runtime/test_multitensor.py	<1MB	bb26d90b775a4455d746b254
test/runtime/test_nn.py	<1MB	b49b4e1cc645ef3868534544
test/runtime/test_ops.py	<1MB	937231c24d02beb61a1c87c8
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test/runtime/test_optim.py	<1MB	c1a05f2477feb1513195351a
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test/runtime/test_search.py	<1KB	85101f05a5cd0ae468b01cf9
test/runtime/test_setitem.py	<1MB	71f9d660c1d683ea19847b9a
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test/testextra/__init__.py	<1KB	3eb88b536f10500edb3c5676

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test/testextra/test_f16_decompress.py	<1KB	99be64028f053f30f9bb769e
test/testextra/test_fp8_linear.py	<1MB	b895c322224a0fd3f4395219
test/testextra/test_hevc.py	<1MB	f9f5a8eaf3af04d530e0aa61
test/testextra/test_hk_fa.py	<1MB	17288fb2734d95ce03bf078c
test/testextra/test_lr_scheduler.py	<1MB	3eeada068873a869797ebb2f
test/testextra/test_mockgpu.py	<1KB	3332c0447d126dea49a1c00f
test/testextra/test_tk.py	<1MB	0aa913ec93091ab34ee5961d
test/web/test_webgpu.js	<1MB	5e4bfdb973478158b8e5a655

Activity: 1148 files read, 0 files written, 49 subprocesses. Full list in Appendix B.

6 Verification

Field	Value
Signer	Signed by the K-Veritas server
Signed at	2026-10-07T07:43:17.469358+00:00
Algorithm	RSA-PSS-SHA256, 4096-bit key
Key fingerprint	eeac81653cf44ac029c4069cf2a06d9f
Data hash	7a4a9afa73e75c11dd8814c9f98bd317f848d771a8272d361e229ef70aa583e7
Code bundle hash	bae3e78018887e89e095c6f54251e33fe7e98e443178b2ea040bb18be22af6a1

Run anchors: 1 invocations were anchored at the server when they ended.

Checks a verifier runs

1. The stored canonical data hashes to the data hash above, and the signature verifies over data_hash:nonce:signed_at.
2. The signing key is the K-Veritas server key, not one supplied by the author.
3. Nothing was appended after the seal, and these pages hash to the value bound in the seal.
4. Each run matches the digest the server received when it ended.
5. A supplied code bundle hashes to the code bundle hash above.

References

- [1] M. K. Keita and C. Homan. Computer Science Conferences Should Require Nonrepudiable Experimental Results. NeurIPS 2026 Position Paper Track. arXiv:2605.08586.

Appendix

A Full metric series

Run 1: kv_run.sh

Metric	Value	Step	Line	Source
cifar_train_loss	1184.55	0	5	explicit
cifar_train_loss	798.95	50	56	explicit
cifar_train_loss	682.53	100	108	explicit
cifar_train_loss	666.02	150	159	explicit
cifar_train_loss	650.83	200	211	explicit
cifar_train_loss	643.1	250	262	explicit
cifar_train_loss	608.68	300	314	explicit
cifar_train_loss	605.56	350	365	explicit
cifar_train_loss	581.06	400	417	explicit
cifar_train_loss	595.05	450	468	explicit
cifar_train_loss	563.98	500	520	explicit
cifar_train_loss	579.64	550	571	explicit
cifar_train_loss	562.68	600	623	explicit
cifar_train_loss	574.45	650	674	explicit
cifar_train_loss	557.76	700	726	explicit
cifar_train_loss	563.97	750	777	explicit
cifar_train_loss	541.23	800	829	explicit
cifar_train_loss	532.05	850	880	explicit
cifar_train_loss	511.12	900	932	explicit
cifar_train_loss	508.86	950	983	explicit
cifar_eval_acc_pct	93.75	1000	1036	explicit
mnist_final_loss	0.08		1112	explicit

Phase	Line	Time	CPU time	Memory	GPU
h1b_cifar10	2	07:37:17	1.6 s	0.12 GB	
beautiful_mnist	1038	07:43:03	0.1 s	0.04 GB	

B Provenance timeline and activity map

Run 1: kv_run.sh

#	Snapshot	Time	Root	Changed
0	run_start	07:37:17	3e0eb97765d9d1b1	1147
1	phase h1b_cifar10	07:37:17	3e0eb97765d9d1b1	0
2	phase beautiful_mnist	07:43:03	3e0eb97765d9d1b1	0
3	run_end	07:43:14	3e0eb97765d9d1b1	0

Event	Path or command	Hash
read	kv_run.sh	
read	examples/h1b_cifar10.py	
read	extra/lr_scheduler.py	
read	tinygrad/__init__.py	
read	tinygrad/tensor.py	
read	tinygrad/runtime/support/__init__.py	

Event	Path or command	Hash
read	tinygrad/mixin/__init__.py	
read	tinygrad/mixin/op.py	
read	tinygrad/mixin/elementwise.py	
read	tinygrad/mixin/creation.py	
read	tinygrad/mixin/reduce.py	
read	tinygrad/schedule/__init__.py	
read	tinygrad/uop/spec.py	
read	tinygrad/uop/render.py	
read	tinygrad/schedule/memory.py	
read	tinygrad/engine/realize.py	
read	tinygrad/renderer/tc.py	
read	tinygrad/renderer/isa/__init__.py	
read	tinygrad/codegen/gpudims.py	
read	tinygrad/codegen/late/linearizer.py	
read	tinygrad/uop/movement.py	
read	tinygrad/codegen/decomp/__init__.py	
read	tinygrad/codegen/decomp/transcendental.py	
read	tinygrad/codegen/opt/ptrange.py	
read	tinygrad/codegen/simplify.py	
read	tinygrad/schedule/multi.py	
read	tinygrad/schedule/indexing.py	
read	tinygrad/codegen/late/regalloc.py	
read	tinygrad/runtime/support/hcq2.py	
read	tinygrad/engine/jit.py	
read	tinygrad/nn/__init__.py	
read	tinygrad/nn/optim.py	
read	tinygrad/nn/state.py	
read	extra/bench_log.py	
read	tinygrad/renderer/llvmir.py	
read	tinygrad/runtime/autogen/kfd.py	
read	tinygrad/runtime/support/c.py	
read	tinygrad/runtime/autogen/hsa.py	
read	tinygrad/runtime/autogen/amdgpu_drm.py	
read	tinygrad/runtime/autogen/am/__init__.py	
read	tinygrad/runtime/autogen/am/am.py	
read	tinygrad/runtime/support/elf.py	
read	tinygrad/runtime/autogen/libc.py	
read	tinygrad/runtime/support/system.py	
read	tinygrad/runtime/autogen/pci.py	
read	tinygrad/runtime/autogen/vfio.py	
read	tinygrad/runtime/support/usb.py	
read	tinygrad/runtime/autogen/libusb.py	
read	tinygrad/runtime/support/am/amdev.py	
read	tinygrad/runtime/support/am/ip.py	
read	.kveritasignore	
read	tinygrad/renderer/ptx.py	
read	.github/actions/process-replay/action.yml	
read	.github/actions/setup-tinygrad/action.yml	
read	.github/workflows/autogen.yml	
read	.github/workflows/benchmark.yml	

Event	Path or command	Hash
read	.github/workflows/benchmark_search.yml	
read	.github/workflows/docs.yml	
read	.github/workflows/mlperf.yml	
read	.github/workflows/platform.yml	
read	.github/workflows/python-publish.yml	
read	.github/workflows/szdiff.yml	
read	.github/workflows/test.yml	
read	.pre-commit-config.yaml	
read	conftest.py	
read	docs/abstractions3.py	
read	docs/abstractions4.py	
read	examples/__init__.py	
read	examples/audio_helpers.py	
read	examples/beautiful_cartpole.py	
read	examples/beautiful_cifar.py	
read	examples/beautiful_mnist.py	
read	examples/beautiful_mnist_multigpu.py	
read	examples/benchmark_onnx.py	
read	examples/compile_efficientnet.py	
read	examples/compile_tensorflow.py	
read	examples/conversation_data/pre_prompt_gary.yaml	
read	examples/conversation_data/pre_prompt_george.yaml	
read	examples/conversation_data/pre_prompt_lexie.yaml	
read	examples/conversation_data/pre_prompt_stacy.yaml	
read	examples/gpt2.py	
read	examples/gradaccum_mnist.py	
read	examples/llama.py	
read	examples/llama3.py	
read	examples/llm.c/export.py	
read	examples/llm.c/train_gpt2.py	
read	examples/llm.c/ubench/matmul.c	
read	examples/mamba.py	
read	examples/minrf.py	
read	examples/mixtral.py	
read	examples/mlperf/dataloader.py	
read	examples/mlperf/helpers.py	
read	examples/mlperf/initializers.py	
read	examples/mlperf/losses.py	
read	examples/mlperf/lr_schedulers.py	
read	examples/mlperf/metrics.py	
read	examples/mlperf/model_eval.py	
read	examples/mlperf/model_spec.py	
read	examples/mlperf/model_train.py	
read	examples/mlperf/models/flat_llama.py	
read	examples/mlperf/models/gpt_oss.py	
read	examples/mlperf/models/test_apply_grad.py	
read	examples/mlperf/models/test_flat_llama.py	
read	examples/mlperf/optim.py	
read	examples/mlperf/scripts/setup_kits19_dataset.sh	
read	examples/mlperf/scripts/stable_diffusion_downloads.sh	

Event	Path or command	Hash
read	examples/mlperf/training_submission_v6.0/tinycorp/benchmarks/llama31_8b/...	
read	examples/mlperf/training_submission_v6.0/tinycorp/benchmarks/resnet/imple...	
read	examples/mlperf/training_submission_v6.0/tinycorp/benchmarks/resnet/imple...	
read	examples/mlperf/training_submission_v6.0/tinycorp/benchmarks/resnet/imple...	
read	examples/mlperf/training_submission_v6.0/tinycorp/benchmarks/resnet/imple...	
read	examples/mlperf/training_submission_v6.0/tinycorp/benchmarks/resnet/imple...	
read	examples/mlperf/training_submission_v6.0/tinycorp/benchmarks/resnet/imple...	
read	examples/mlperf/training_submission_v6.0/tinycorp/benchmarks/resnet/imple...	
read	examples/mlperf/training_submission_v6.0/tinycorp/benchmarks/retinanet/im...	
read	examples/mlperf/training_submission_v6.0/tinycorp/benchmarks/retinanet/im...	
read	examples/mlperf/training_submission_v6.0/tinycorp/benchmarks/retinanet/im...	
read	examples/mlperf/training_submission_v6.0/tinycorp/benchmarks/retinanet/im...	
read	examples/mlperf/training_submission_v6.0/tinycorp/benchmarks/retinanet/im...	
read	examples/mlperf/training_submission_v6.0/tinycorp/systems/tinybox_8xMI3...	
read	examples/mlperf/training_submission_v6.0/tinycorp/systems/tinybox_8xMI3...	
read	examples/mlperf/training_submission_v6.0/tinycorp/systems/tinybox_green.j...	
read	examples/mlperf/training_submission_v6.0/tinycorp/systems/tinybox_red.json	
read	examples/mlperf/training_submission_v6.1/tinycorp/benchmarks/gpt_oss/imp...	
read	examples/mlperf/training_submission_v6.1/tinycorp/benchmarks/gpt_oss/imp...	
read	examples/mlperf/training_submission_v6.1/tinycorp/benchmarks/llama31_8b/...	
read	examples/mlperf/training_submission_v6.1/tinycorp/benchmarks/llama31_8b/...	
read	examples/mlperf/training_submission_v6.1/tinycorp/benchmarks/llama31_8b/...	
read	examples/mlperf/training_submission_v6.1/tinycorp/benchmarks/llama31_8b/...	
read	examples/mlperf/training_submission_v6.1/tinycorp/benchmarks/llama31_8b/...	
read	examples/mlperf/training_submission_v6.1/tinycorp/benchmarks/llama31_8b/...	
read	examples/mlperf/training_submission_v6.1/tinycorp/benchmarks/llama31_8b/...	
read	examples/mlperf/training_submission_v6.1/tinycorp/benchmarks/llama31_8b/...	
read	examples/mlperf/training_submission_v6.1/tinycorp/systems/tinybox_8xMI3...	
read	examples/mlperf/training_submission_v6.1/tinycorp/systems/tinybox_8xMI3...	
read	examples/mlperf/training_submission_v6.1/tinycorp/systems/tinybox_green.j...	
read	examples/mlperf/training_submission_v6.1/tinycorp/systems/tinybox_red.json	
read	examples/mnist_gan.py	
read	examples/olmoe.py	
read	examples/openpilot/check_image_reads.py	
read	examples/openpilot/compile_onnx.py	
read	examples/openpilot/compile_warp.py	
read	examples/openpilot/helpers.py	
read	examples/openpilot/load_pickle.py	
read	examples/other_mnist/beautiful_mnist_torch.py	
read	examples/qwq.py	
read	examples/sdv2.py	
read	examples/sd-xl.py	
read	examples/stable_diffusion.py	
read	examples/stunning_mnist.py	
read	examples/test_onnx_imagenet.py	
read	examples/test_pkl_imagenet.py	
read	examples/tinychat/assets/unpkg.com/@highlightjs/cdn-assets@11.9.0/highlig...	
read	examples/tinychat/assets/unpkg.com/marked-highlight@2.1.2/lib/index.umd.js	

Event	Path or command	Hash
read	examples/tinychat/assets/unpkg.com/marked@13.0.0/marked.min.js	
read	examples/tinychat/fetch_assets.sh	
read	examples/tinychat/index.js	
read	examples/tinychat/tinychat-browser/compile.py	
read	examples/tinychat/tinychat-browser/compile_wasm.sh	
read	examples/tinychat/tinychat-browser/index.js	
read	examples/tinychat/tinychat-browser/make_tiktoken_js.sh	
read	examples/tinychat/tinychat-browser/tiktoken-export.js	
read	examples/tinychat/tinychat-browser/webpack.config.js	
read	examples/tinychat/tinychat-browser/worker.js	
read	examples/tools/bandwidth_test.py	
read	examples/tools/gpuburn.py	
read	examples/torch_cuda_kernel.py	
read	examples/train_resnet.py	
read	examples/transformer.py	
read	examples/vgg7.py	
read	examples/vgg7_helpers/waifu2x.py	
read	examples/webgpu/stable_diffusion/compile.py	
read	examples/webgpu/yolov8/compile.py	
read	examples/whisper.py	
read	examples/yolov3.py	
read	examples/yolov8-onnx.py	
read	examples/yolov8.py	
read	extra/amdflash/common.py	
read	extra/amdflash/fw_live.py	
read	extra/amdflash/romless.py	
read	extra/amdpci/am_smi.py	
read	extra/amdpci/headers/amdgpu_doorbell.h	
read	extra/amdpci/headers/amdgpu_irq.h	
read	extra/amdpci/headers/amdgpu_psp.h	
read	extra/amdpci/headers/amdgpu_smu.h	
read	extra/amdpci/headers/amdgpu_ucode.h	
read	extra/amdpci/headers/amdgpu_vm.h	
read	extra/amdpci/headers/discovery.h	
read	extra/amdpci/headers/psp_gfx_if.h	
read	extra/amdpci/headers/v11_structs.h	
read	extra/amdpci/headers/v12_structs.h	
read	extra/amdpci/hive_reset.py	
read	extra/amdpci/overlay/gc_9_4_3.h	
read	extra/amdpci/proclogs.py	
read	extra/amdpci/setup_python_cap.sh	
read	extra/amdpci/setup_vfio.sh	
read	extra/archprobe.py	
read	extra/benchmark_llm.py	
read	extra/bnxt_driver/connect.py	
read	extra/bnxt_driver/loopback.py	
read	extra/bnxt_driver/test_bnxt.py	
read	extra/bnxt_driver/test_bnxt_transport.py	
read	extra/cl_android.sh	
read	extra/datasets/__init__.py	

Event	Path or command	Hash
read	extra/datasets/fake_imagenet_from_mnist.py	
read	extra/datasets/imagenet.py	
read	extra/datasets/imagenet_download.py	
read	extra/datasets/kits19.py	
read	extra/datasets/librispeech.py	
read	extra/datasets/openimages.py	
read	extra/datasets/preprocess_imagenet.py	
read	extra/datasets/squad.py	
read	extra/datasets/wikipedia.py	
read	extra/datasets/wikipedia_download.py	
read	extra/dsp/compile.py	
read	extra/dsp/gen.sh	
read	extra/dsp/hook.py	
read	extra/dsp/include/AEEStdDef.h	
read	extra/dsp/include/AEEStdErr.h	
read	extra/dsp/include/HAP_power.h	
read	extra/dsp/include/adsprpc_shared.h	
read	extra/dsp/include/apps_std.h	
read	extra/dsp/include/ion.h	
read	extra/dsp/include/msm_ion.h	
read	extra/dsp/include/remote_default.h	
read	extra/dsp/invoke_bug.py	
read	extra/dsp/invoke_bug_2.py	
read	extra/dsp/opt.py	
read	extra/dsp/preload.c	
read	extra/dsp/run.py	
read	extra/dsp/run_3.py	
read	extra/dsp/snpe.sh	
read	extra/dsp/snpe_logs/parse.py	
read	extra/export_model.py	
read	extra/f16_decompress.py	
read	extra/fp8/fp8_linear.py	
read	extra/gemm/amd_asm_matmul.py	
read	extra/gemm/amd_call_matmul.py	
read	extra/gemm/amd_copy_matmul.py	
read	extra/gemm/amd_matmul.py	
read	extra/gemm/amd_seb/kernel3_registers.cpp	
read	extra/gemm/amd_seb/kernel4_gmem_df.cpp	
read	extra/gemm/amd_seb/kernel5_lds_optim.cpp	
read	extra/gemm/amd_uop_matmul.py	
read	extra/gemm/cdna_asm_gemm.py	
read	extra/gemm/cuda_matmul.py	
read	extra/gemm/fuzz_matmul.py	
read	extra/gemm/gemm.c	
read	extra/gemm/gemm.py	
read	extra/gemm/gemm_mxfp4.py	
read	extra/gemm/halide_gemm.py	
read	extra/gemm/hip_matmul.py	
read	extra/gemm/max_matmul.py	
read	extra/gemm/metal_conv.py	

Event	Path or command	Hash
read	extra/gemm/metal_matmul.py	
read	extra/gemm/metal_matvec.py	
read	extra/gemm/metal_uop_matmul.py	
read	extra/gemm/mi350x_uop_matmul.py	
read	extra/gemm/mi350x_uop_matmul_2.py	
read	extra/gemm/moe_gemm.py	
read	extra/gemm/moe_routing.py	
read	extra/gemm/rdna4_asm_matmul.py	
read	extra/gemm/real_pmatmul.py	
read	extra/gemm/simple_conv.py	
read	extra/gemm/simple_matmul.py	
read	extra/gemm/simple_matvec.py	
read	extra/gemm/tinygrad_nv_matmul.py	
read	extra/gemm/torch_gemm.py	
read	extra/gemm/triton_nv_matmul.py	
read	extra/gemm/tvm_gemm.py	
read	extra/gemm_fragment.py	
read	extra/gptoss_kernels/combine/__init__.py	
read	extra/gptoss_kernels/combine/forward.cpp	
read	extra/gptoss_kernels/combine/weights_backward.cpp	
read	extra/gptoss_kernels/dispatch/__init__.py	
read	extra/gptoss_kernels/dispatch/gather.cpp	
read	extra/gptoss_kernels/dispatch/inverse.cpp	
read	extra/gptoss_kernels/embedding/__init__.py	
read	extra/gptoss_kernels/embedding/embedding_bwd.cpp	
read	extra/gptoss_kernels/embedding/embedding_fwd.cpp	
read	extra/gptoss_kernels/gather_sum/__init__.py	
read	extra/gptoss_kernels/gather_sum/gather_sum.cpp	
read	extra/gptoss_kernels/grad_norm/__init__.py	
read	extra/gptoss_kernels/grad_norm/sum_squares.cpp	
read	extra/gptoss_kernels/qkv_rope/__init__.py	
read	extra/gptoss_kernels/qkv_rope/backward.cpp	
read	extra/gptoss_kernels/qkv_rope/forward.cpp	
read	extra/gptoss_kernels/quantize_mxfp8/__init__.py	
read	extra/gptoss_kernels/residual/__init__.py	
read	extra/gptoss_kernels/residual/residual.cpp	
read	extra/gptoss_kernels/residual/wo_bias_final.cpp	
read	extra/gptoss_kernels/residual/wo_bias_partial.cpp	
read	extra/gptoss_kernels/rmsnorm/__init__.py	
read	extra/gptoss_kernels/rmsnorm/fast_final_denom.cpp	
read	extra/gptoss_kernels/rmsnorm/rmsnorm_mul_bwd.cpp	
read	extra/gptoss_kernels/rmsnorm/rmsnorm_mul_quantize_mxfp8.cpp	
read	extra/gptoss_kernels/rmsnorm/rmsnorm_mul_quantize_mxfp8_bwd.cpp	
read	extra/gptoss_kernels/router_topk/__init__.py	
read	extra/gptoss_kernels/router_topk/backward.cpp	
read	extra/gptoss_kernels/router_topk/forward.cpp	
read	extra/gptoss_kernels/router_topk/input_backward.cpp	
read	extra/gptoss_kernels/router_topk/weight_backward.cpp	
read	extra/gradcheck.py	
read	extra/hcq/hcq_smi.py	

Event	Path or command	Hash
read	extra/hcq1/__init__.py	
read	extra/hcq1/graph.py	
read	extra/hcq1/hcq.py	
read	extra/hcq1/ops_amd_old.py	
read	extra/hcq1/ops_rdma.py	
read	extra/hcq1/remote.py	
read	extra/hcq1/test_hcq.py	
read	extra/hcq1/test_hcq_graph.py	
read	extra/hcqfuzz/fuzzer.py	
read	extra/hcqfuzz/spec.py	
read	extra/hcqfuzz/tests/allocator.py	
read	extra/hcqfuzz/tests/allreduce.py	
read	extra/hcqfuzz/tests/bert.py	
read	extra/hcqfuzz/tests/hcq.py	
read	extra/hcqfuzz/tests/resnet.py	
read	extra/hcqfuzz/tools.py	
read	extra/hevc/decode.py	
read	extra/hevc/hevc.py	
read	extra/hip_gpu_driver/amdgpu_drm.h	
read	extra/hip_gpu_driver/gc_10_3_0_offset.h	
read	tinygrad/runtime/autogen/nv_610.py	
read	extra/hip_gpu_driver/gc_11_0_0_offset.h	
read	extra/hip_gpu_driver/hip_ioctl.py	
read	extra/hip_gpu_driver/kfd_ioctl.h	
read	extra/hip_gpu_driver/kfd_pm4_headers_ai.h	
read	extra/hip_gpu_driver/navi10_sdma_pkt_open.h	
read	extra/hip_gpu_driver/nvd.h	
read	extra/hip_gpu_driver/sdma_registers.h	
read	extra/hip_gpu_driver/sdma_v6_0_0_pkt_open.h	
read	extra/hip_gpu_driver/sienna_cichlid_ip_offset.h	
read	extra/hip_gpu_driver/soc15d.h	
read	extra/hip_gpu_driver/soc21_enum.h	
read	extra/hip_gpu_driver/test_kfd_2.py	
read	extra/hip_gpu_driver/test_pm4.py	
read	extra/hip_gpu_driver/test_sdma_fun.py	
read	extra/hip_gpu_driver/vega10_sdma_pkt_open.h	
read	extra/hip_large_kernel.py	
read	extra/hiprtc/hiprtc_runtime.h	
read	extra/hook_cuda.py	
read	extra/huggingface_onnx/huggingface_manager.py	
read	extra/huggingface_onnx/run_models.py	
read	extra/introspection.py	
read	extra/llama_kernels/__init__.py	
read	extra/llama_kernels/fused_ce/__init__.py	
read	extra/llama_kernels/quantize_mxfp4/__init__.py	
read	extra/llama_kernels/quantize_mxfp4/quantize_mxfp4.cpp	
read	extra/llama_kernels/quantize_mxfp4/quantize_mxfp4_device.h	
read	extra/llama_kernels/rmsnorm/__init__.py	
read	extra/llama_kernels/swiglu/__init__.py	
read	extra/llama_kernels/swiglu/swiglu_bwd_mxfp4.cpp	

Event	Path or command	Hash
read	extra/llama_kernels/swiglu/swiglu_fwd_mxfp4.cpp	
read	extra/llama_kernels/transpose_quantize_mxfp8/__init__.py	
read	extra/llama_kernels/transpose_quantize_mxfp8/transpose_quantize_mxfp8.cpp	
read	extra/mesa/lvp_nir_options.sh	
read	extra/mlx_driver/connect.py	
read	extra/mlx_driver/loopback.py	
read	extra/mlx_driver/mlx5.h	
read	extra/mlx_driver/mlxdev.py	
read	extra/mmapeak/mmapeak.py	
read	extra/models/bert.py	
read	extra/models/clip.py	
read	extra/models/convnext.py	
read	extra/models/efficientnet.py	
read	extra/models/inception.py	
read	extra/models/llama.py	
read	extra/models/mask_rcnn.py	
read	extra/models/resnet.py	
read	extra/models/retinanet.py	
read	extra/models/rnnt.py	
read	extra/models/t5.py	
read	extra/models/transformer.py	
read	extra/models/unet.py	
read	extra/models/unet3d.py	
read	extra/models/vit.py	
read	extra/multitensor.py	
read	extra/nvJitLink.h	
read	extra/nv_gpu_driver/clc6c0qmd.h	
read	extra/nv_gpu_driver/clc9b0.h	
read	extra/nv_gpu_driver/clcec0qmd.h	
read	extra/nv_gpu_driver/fw.h	
read	extra/nv_gpu_driver/g_rpc-message-header.h	
read	extra/nv_gpu_driver/gsp_static_config.h	
read	extra/nv_gpu_driver/hs.h	
read	extra/nv_gpu_driver/nv_ioctl.py	
read	extra/nv_gpu_driver/nvdec_drv.h	
read	extra/nv_gpu_driver/pci_exp_table.h	
read	extra/nv_gpu_driver/vbios.h	
read	extra/nv_pma/collect.py	
read	extra/nv_pma/cupti/__init__.py	
read	extra/nv_pma/cupti/cu_prof_ext.py	
read	extra/nv_pma/cupti/cupti.py	
read	extra/nv_pma/decode.py	
read	extra/nv_pma/test/test_nvprof.py	
read	extra/onnx_helpers.py	
read	extra/optimization/extract_dataset.py	
read	extra/optimization/generate_dataset.sh	
read	extra/optimization/test_beam_search.py	
read	extra/perfetto/to_perfetto.py	
read	extra/qcom_gpu_driver/a6xx.xml.h	
read	extra/qcom_gpu_driver/gen.sh	

Event	Path or command	Hash
read	extra/qcom_gpu_driver/msm_kgsl.h	
read	extra/qcom_gpu_driver/msm_kgsl.py	
read	extra/qcom_gpu_driver/openccl_ioctl.py	
read	extra/qcom_gpu_driver/qcom_openccl_interop.py	
read	extra/remote/bench.py	
read	extra/remote/serve.py	
read	extra/setup_hipcc_osx.sh	
read	extra/setup_hipcomgr_osx.sh	
read	extra/setup_mock_amd_osx.sh	
read	extra/setup_mock_dsp_osx.sh	
read	extra/setup_mock_nv_osx.sh	
read	extra/setup_nvcc_osx.sh	
read	extra/setup_tinygpu_osx.sh	
read	extra/sqtt/examples/generate_examples.py	
read	extra/sqtt/install_rocprof_decoder.py	
read	extra/sqtt/rgptool.py	
read	extra/sqtt/roc.py	
read	extra/sqtt/sqtt.h	
read	extra/test_mi350.sh	
read	extra/testsig/generate_testsig.py	
read	extra/thneed.py	
read	extra/thunder/amd/fa.py	
read	extra/thunder/amd/fa_bwd_causal.cpp	
read	extra/thunder/amd/fa_bwd_post.cpp	
read	extra/thunder/amd/fa_bwd_pre.cpp	
read	extra/thunder/amd/fa_fwd_causal.cpp	
read	extra/thunder/amd/fused_qkv_rope_bwd.cpp	
read	extra/thunder/amd/gemm_bf16.cpp	
read	extra/thunder/amd/gemm_bf16_atb.cpp	
read	extra/thunder/amd/gemm_mxfp8.cpp	
read	extra/thunder/amd/grouped_mxfp8_gemm.cpp	
read	extra/thunder/amd/grouped_mxfp8_wgrad.cpp	
read	extra/thunder/amd/include/utils.cpp	
read	extra/thunder/amd/moe_router_mfma.cpp	
read	extra/thunder/cuda/fa.py	
read	extra/thunder/cuda/matmul.py	
read	extra/thunder/metal/gemm.py	
read	extra/thunder/tiny/fa.py	
read	extra/thunder/tiny/tk/__init__.py	
read	extra/thunder/tiny/tk/group.py	
read	extra/thunder/tiny/tk/kernel.py	
read	extra/thunder/tiny/tk/tiles.py	
read	extra/thunder/tiny/visualize_tile.py	
read	extra/tinydreno.h	
read	extra/torch_backend/__init__.py	
read	extra/torch_backend/backend.py	
read	extra/torch_backend/backend2.py	
read	extra/torch_backend/example.py	
read	extra/torch_backend/test.py	
read	extra/torch_backend/test_compile.py	

Event	Path or command	Hash
read	extra/torch_backend/test_inplace.py	
read	extra/torch_backend/test_kernel_fusion.py	
read	extra/torch_backend/test_multigpu.py	
read	extra/torch_backend/torch_tests.py	
read	extra/torch_backend/wrapped_tensor.cpp	
read	extra/torch_hook/hook_cuda.py	
read	extra/torch_hook/hook_torch.py	
read	extra/training.py	
read	extra/usbgpu/debug.py	
read	extra/usbgpu/legacy/nvme_speed.py	
read	extra/usbgpu/legacy/patch_exp.py	
read	extra/usbgpu/legacy/wr_speed.py	
read	extra/usbgpu/patch.py	
read	extra/usbgpu/scan_pci.py	
read	extra/usbgpu/tbgpu/install_tinygpu.sh	
read	extra/usbgpu/tbgpu/installer/Shared/Assets.xcassets/AccentColor.colorset/Co...	
read	extra/usbgpu/tbgpu/installer/Shared/Assets.xcassets/AppIcon.appiconset/Cont...	
read	extra/usbgpu/tbgpu/installer/Shared/Assets.xcassets/Contents.json	
read	extra/usbgpu/tbgpu/installer/Shared/TinyGPU-Bridging-Header.h	
read	extra/usbgpu/tbgpu/installer/Shared/server.c	
read	extra/usbgpu/tbgpu/installer/Shared/tiny_icon.icon/icon.json	
read	extra/usbgpu/tbgpu/installer/TinyGPUDriverExtension/TinyGPUDriver.cpp	
read	extra/usbgpu/tbgpu/installer/TinyGPUDriverExtension/TinyGPUDriverUserC...	
read	extra/usbgpu/tbgpu/installer/build_and_sign.sh	
read	extra/usbgpu/tbgpu/installer/build_and_sign_nv.sh	
read	extra/usbgpu/tbgpu/installer/install_nosip.sh	
read	extra/usbgpu/tbgpu/installer/notary_tool.sh	
read	extra/usbgpu/tbgpu/kill_tinygpu.sh	
read	extra/webgpu/webgpu.h	
read	extra/weekly_commits_table.py	
read	mkdocs.yml	
read	opencode.json	
read	pyproject.toml	
read	serve_docs.sh	
read	spec/render.sh	
read	sz.py	
read	test/__init__.py	
read	test/amd/__init__.py	
read	test/amd/disasm.py	
read	test/amd/helpers.py	
read	test/amd/hw/__init__.py	
read	test/amd/hw/helpers.py	
read	test/amd/hw/test_cdna_sdwa.py	
read	test/amd/hw/test_cdna_vop3.py	
read	test/amd/hw/test_dpp.py	
read	test/amd/hw/test_ds.py	
read	test/amd/hw/test_flat.py	
read	test/amd/hw/test_global.py	
read	test/amd/hw/test_rdna4_fp8.py	
read	test/amd/hw/test_rdna4_permlane_var.py	

Event	Path or command	Hash
read	test/amd/hw/test_scratch.py	
read	test/amd/hw/test_smem.py	
read	test/amd/hw/test_sop.py	
read	test/amd/hw/test_vinterp.py	
read	test/amd/hw/test_vop1.py	
read	test/amd/hw/test_vop2.py	
read	test/amd/hw/test_vop3.py	
read	test/amd/hw/test_vop3p.py	
read	test/amd/hw/test_vopc.py	
read	test/amd/hw/test_vopd.py	
read	test/amd/test_asm_kernel.py	
read	test/amd/test_dsl2.py	
read	test/amd/test_emu2_pcode.py	
read	test/amd/test_emu_memory.py	
read	test/amd/test_formats.py	
read	test/amd/test_handwritten.py	
read	test/amd/test_integration.py	
read	test/amd/test_llvm.py	
read	test/amd/test_mockgpu_invalid.py	
read	test/amd/test_pdf.py	
read	test/amd/test_rdna3_asm.py	
read	test/amd/test_roundtrip.py	
read	test/amd/test_sqtt_encoder.py	
read	test/amd/test_sqtt_examples.py	
read	test/amd/test_sqtt_profiler.py	
read	test/amd/test_sqttmap.py	
read	test/device/__init__.py	
read	test/device/amd/__init__.py	
read	test/device/amd/test_arange.py	
read	test/device/amd/test_asm_gemm.py	
read	test/device/amd/test_attention.py	
read	test/device/amd/test_llama_kernels.py	
read	test/device/amd/test_llm.py	
read	test/device/amd/test_llvm.py	
read	test/device/cl/__init__.py	
read	test/device/cl/test_ocl.py	
read	test/device/cpu/__init__.py	
read	test/device/cpu/test_call.py	
read	test/device/cpu/test_cpu.py	
read	test/device/cpu/test_custom_kernel.py	
read	test/device/cpu/test_ops.py	
read	test/device/cpu/test_quantize_onnx.py	
read	test/device/dsp/__init__.py	
read	test/device/dsp/test_quantize_onnx.py	
read	test/device/metal/__init__.py	
read	test/device/metal/test_metal.py	
read	test/device/metal/test_softmax_fusion.py	
read	test/device/nv/__init__.py	
read	test/device/nv/test_renderer_failures.py	
read	test/device/python/__init__.py	

Event	Path or command	Hash
read	test/device/python/test_uops.py	
read	test/device/qcom/__init__.py	
read	test/device/qcom/test_qcom.py	
read	test/device/webgpu/__init__.py	
read	test/device/webgpu/test_renderer_failures.py	
read	test/device/webgpu/test_uops.py	
read	test/external/external_benchmark_all2all.py	
read	test/external/external_benchmark_am.py	
read	test/external/external_benchmark_bert_matmuls.py	
read	test/external/external_benchmark_bert_softmax.py	
read	test/external/external_benchmark_disk_raw.py	
read	test/external/external_benchmark_hip_compile.py	
read	test/external/external_benchmark_keccak.py	
read	test/external/external_benchmark_kernel_launch.py	
read	test/external/external_benchmark_llama_offload.py	
read	test/external/external_benchmark_llama_schedule.py	
read	test/external/external_benchmark_multitensor_allreduce.py	
read	test/external/external_benchmark_op_conv.py	
read	test/external/external_benchmark_resnet.py	
read	test/external/external_benchmark_schedule.py	
read	test/external/external_cl_half_max.py	
read	test/external/external_fuzz_ampt.py	
read	test/external/external_fuzz_beam_timeout_recovery.py	
read	test/external/external_fuzz_hcq_mp.py	
read	test/external/external_fuzz_hcq_signals.py	
read	test/external/external_fuzz_qcom_cpu_cache.py	
read	test/external/external_fuzz_sdma_warm_start.py	
read	test/external/external_fuzz_tlsf.py	
read	test/external/external_gpu_fail_osx.py	
read	test/external/external_jit_failure.py	
read	test/external/external_llm_eval.py	
read	test/external/external_metal_compile_fail.py	
read	test/external/external_model_benchmark.py	
read	test/external/external_multi_gpu.py	
read	test/external/external_nhwc_conv.py	
read	test/external/external_openpilot_image_warp.py	
read	test/external/external_osx_profiling.py	
read	test/external/external_test_am.py	
read	test/external/external_test_am_fault_recovery.py	
read	test/external/external_test_amd.py	
read	test/external/external_test_compile.py	
read	test/external/external_test_datasets.py	
read	test/external/external_test_dev_var.py	
read	test/external/external_test_example.py	
read	test/external/external_test_gpu_crash.py	
read	test/external/external_test_hcq.py	
read	test/external/external_test_hcq_open.py	
read	test/external/external_test_hip_compile.py	
read	test/external/external_test_hive_reset.py	
read	test/external/external_test_image.py	

Event	Path or command	Hash
read	test/external/external_test_jit_on_models.py	
read	test/external/external_test_keccak.py	
read	test/external/external_test_llama3_ff.py	
read	test/external/external_test_llama3_layer.py	
read	test/external/external_test_lm_head.py	
read	test/external/external_test_losses.py	
read	test/external/external_test_lr_schedule.py	
read	test/external/external_test_mamba.py	
read	test/external/external_test_metrics.py	
read	test/external/external_test_mnist_data_select.py	
read	test/external/external_test_onnx_backend.py	
read	test/external/external_test_onnx_ops.py	
read	test/external/external_test_onnx_runner.py	
read	test/external/external_test_opt.py	
read	test/external/external_test_optim.py	
read	test/external/external_test_process_count.py	
read	test/external/external_test_schedule_scaling.py	
read	test/external/external_test_simple_tokenizer.py	
read	test/external/external_test_speed_llama.py	
read	test/external/external_test_speed_theoretical.py	
read	test/external/external_test_tlsf.py	
read	test/external/external_test_usb_asm24.py	
read	test/external/external_test_whisper_librispeech.py	
read	test/external/external_test_yolov8.py	
read	test/external/external_uop_gc.py	
read	test/external/fuzz_fast_idiv.py	
read	test/external/fuzz_graph.py	
read	test/external/fuzz_kfd.py	
read	test/external/fuzz_shape_ops.py	
read	test/external/fuzz_symbolic.py	
read	test/external/fuzz_symbolic_div.py	
read	test/external/mlperf_bert/external_benchmark_bert.py	
read	test/external/mlperf_bert/external_test_checkpoint_loading.py	
read	test/external/mlperf_bert/preprocessing/create_pretraining_data.py	
read	test/external/mlperf_bert/preprocessing/external_test_preprocessing_part.py	
read	test/external/mlperf_bert/preprocessing/pick_eval_samples.py	
read	test/external/mlperf_bert/preprocessing/tokenization.py	
read	test/external/mlperf_resnet/lars_optimizer.py	
read	test/external/mlperf_resnet/lars_util.py	
read	test/external/mlperf_retinanet/coco_utils.py	
read	test/external/mlperf_retinanet/focal_loss.py	
read	test/external/mlperf_retinanet/model/boxes.py	
read	test/external/mlperf_retinanet/model/image_list.py	
read	test/external/mlperf_retinanet/model/transform.py	
read	test/external/mlperf_retinanet/model/utils.py	
read	test/external/mlperf_retinanet/openimages.py	
read	test/external/mlperf_retinanet/presets.py	
read	test/external/mlperf_retinanet/transforms.py	
read	test/external/mlperf_stable_diffusion/external_test_eval.py	
read	test/external/mlperf_stable_diffusion/external_test_models.py	

Event	Path or command	Hash
read	test/external/mlperf_stable_diffusion/external_test_train.py	
read	test/external/mlperf_unet3d/dice.py	
read	test/external/mlperf_unet3d/kits19.py	
read	test/external/process_replay/local.sh	
read	test/external/process_replay/process_replay.py	
read	test/external/process_replay/reset.py	
read	test/external/sglang_llama/external_llama_eval.py	
read	test/external/sglang_llama/mgsm.yaml	
read	test/external/speed_v_theoretical.py	
read	test/helpers.py	
read	test/mockgpu/am/__init__.py	
read	test/mockgpu/am/amdriver.py	
read	test/mockgpu/am/amgpu.py	
read	test/mockgpu/am/amddriver.py	
read	test/mockgpu/am/amdgpu.py	
read	test/mockgpu/am/call.py	
read	test/mockgpu/am/emu.py	
read	test/mockgpu/am/pcode.py	
read	test/mockgpu/am/sqtt_enc.py	
read	test/mockgpu/cuda/cuda.py	
read	test/mockgpu/driver.py	
read	test/mockgpu/gpu.py	
read	test/mockgpu/helpers.py	
read	test/mockgpu/mockgpu.py	
read	test/mockgpu/nv/nvdriver.py	
read	test/mockgpu/nv/nvgpu.py	
read	test/mockgpu/usb.py	
read	test/models/__init__.py	
read	test/models/test_bert.py	
read	test/models/test_efficientnet.py	
read	test/models/test_end2end.py	
read	test/models/test_mnist.py	
read	test/models/test_onnx.py	
read	test/models/test_rnnt.py	
read	test/models/test_train.py	
read	test/models/test_waifu2x.py	
read	test/models/test_whisper.py	
read	test/null/__init__.py	
read	test/null/test_allreduce.py	
read	test/null/test_arange.py	
read	test/null/test_assign.py	
read	test/null/test_attention.py	
read	test/null/test_autogen.py	
read	test/null/test_call.py	
read	test/null/test_compile_failures.py	
read	test/null/test_const_folding.py	
read	test/null/test_custom_kernel.py	
read	test/null/test_device.py	
read	test/null/test_disk_cache.py	
read	test/null/test_dtype.py	

Event	Path or command	Hash
read	test/null/test_dtype_spec.py	
read	test/null/test_dtype_weak.py	
read	test/null/test_edgcases.py	
read	test/null/test_elf.py	
read	test/null/test_encodings.py	
read	test/null/test_function.py	
read	test/null/test_gc.py	
read	test/null/test_gen_float4.py	
read	test/null/test_gguf.py	
read	test/null/test_gpudims.py	
read	test/null/test_gradient.py	
read	test/null/test_graph_rewrite.py	
read	test/null/test_hashing.py	
read	test/null/test_hcq2.py	
read	test/null/test_hcq_iface.py	
read	test/null/test_helpers.py	
read	test/null/test_indexing.py	
read	test/null/test_invalid_tensor.py	
read	test/null/test_isel.py	
read	test/null/test_linearizer.py	
read	test/null/test_linearizer_failures.py	
read	test/null/test_linearizer_rewrite.py	
read	test/null/test_llm_mla.py	
read	test/null/test_llm_server.py	
read	test/null/test_llm_tokenizer.py	
read	test/null/test_memory_planner.py	
read	test/null/test_method_cache.py	
read	test/null/test_microbenchmarks.py	
read	test/null/test_mnist_dataset.py	
read	test/null/test_multitensor.py	
read	test/null/test_nn.py	
read	test/null/test_objc.py	
read	test/null/test_ops.py	
read	test/null/test_pattern_matcher.py	
read	test/null/test_pickle.py	
read	test/null/test_postrange.py	
read	test/null/test_process_replay.py	
read	test/null/test_randomness.py	
read	test/null/test_real_world.py	
read	test/null/test_rearrange_einops.py	
read	test/null/test_renderer_failures.py	
read	test/null/test_resnet.py	
read	test/null/test_rewrite_bottom_up_gate.py	
read	test/null/test_schedule.py	
read	test/null/test_schedule_cache.py	
read	test/null/test_setitem.py	
read	test/null/test_setitem_schedule.py	
read	test/null/test_simplify_valid_idx.py	
read	test/null/test_symbolic_failures.py	
read	test/null/test_symbolic_ops.py	

Event	Path or command	Hash
read	test/null/test_symbolic_tensor.py	
read	test/null/test_system_pci_scan_bus.py	
read	test/null/test_tensor.py	
read	test/null/test_tensor_io.py	
read	test/null/test_tensor_metadata.py	
read	test/null/test_tensor_uop_mixin.py	
read	test/null/test_tensor_uop_representation.py	
read	test/null/test_tensor_variable.py	
read	test/null/test_tqdm.py	
read	test/null/test_transcendental.py	
read	test/null/test_transcendental_helpers.py	
read	test/null/test_uir_render.py	
read	test/null/test_uop_graph.py	
read	test/null/test_uop_repr.py	
read	test/null/test_uop_resolve.py	
read	test/null/test_uop_symbolic.py	
read	test/null/test_uop_vmin_vmax.py	
read	test/null/test_uops.py	
read	test/null/test_uops_stats.py	
read	test/null/test_upat_compile.py	
read	test/null/test_validate_oob.py	
read	test/null/test_viz.py	
read	test/null/test_winograd.py	
read	test/runtime/__init__.py	
read	test/runtime/test_after.py	
read	test/runtime/test_allreduce.py	
read	test/runtime/test_arange.py	
read	test/runtime/test_asm_gemm.py	
read	test/runtime/test_assign.py	
read	test/runtime/test_attention.py	
read	test/runtime/test_buffer.py	
read	test/runtime/test_call.py	
read	test/runtime/test_const_folding.py	
read	test/runtime/test_conv.py	
read	test/runtime/test_custom_kernel.py	
read	test/runtime/test_disk_tensor.py	
read	test/runtime/test_dtype.py	
read	test/runtime/test_dtype_alu.py	
read	test/runtime/test_dtype_spec.py	
read	tinygrad/runtime/support/nv/nvdev.py	
read	test/runtime/test_dtype_weak.py	
read	test/runtime/test_edgecases.py	
read	test/runtime/test_function.py	
read	tinygrad/runtime/support/nv/ip.py	
read	test/runtime/test_getitem_ops.py	
read	test/runtime/test_gguf.py	
read	test/runtime/test_gradient.py	
read	test/runtime/test_hashing.py	
read	test/runtime/test_hcq2.py	
read	test/runtime/test_helpers.py	

Event	Path or command	Hash
read	test/runtime/test_indexing.py	
read	test/runtime/test_interop.py	
read	test/runtime/test_invalid_tensor.py	
read	test/runtime/test_jit.py	
read	test/runtime/test_jit_cases.py	
read	test/runtime/test_jit_footguns.py	
read	test/runtime/test_kernel_opts.py	
read	test/runtime/test_linalg.py	
read	test/runtime/test_linearizer.py	
read	test/runtime/test_linearizer_dumb.py	
read	test/runtime/test_llama_kernels.py	
read	test/runtime/test_llm_mla.py	
read	test/runtime/test_llm_moe.py	
read	test/runtime/test_llm_quantized.py	
read	test/runtime/test_llm_server.py	
read	test/runtime/test_llm_tp.py	
read	test/runtime/test_masked_tensor.py	
read	test/runtime/test_multitensor.py	
read	test/runtime/test_nn.py	
read	test/runtime/test_ops.py	
read	test/runtime/test_opt_gemm.py	
read	test/runtime/test_optim.py	
read	test/runtime/test_pickle.py	
read	test/runtime/test_profiler.py	
read	test/runtime/test_randomness.py	
read	test/runtime/test_rangeify.py	
read	test/runtime/test_realize_is_realize.py	
read	test/runtime/test_rearrange_einops.py	
read	test/runtime/test_renderer_failures.py	
read	test/runtime/test_schedule.py	
read	test/runtime/test_schedule_cache.py	
read	test/runtime/test_search.py	
read	test/runtime/test_setitem.py	
read	test/runtime/test_setitem_schedule.py	
read	test/runtime/test_shm_tensor.py	
read	test/runtime/test_softmax_fusion.py	
read	test/runtime/test_stunning.py	
read	test/runtime/test_subbuffer.py	
read	test/runtime/test_symbolic_jit.py	
read	test/runtime/test_symbolic_ops.py	
read	test/runtime/test_symbolic_tensor.py	
read	test/runtime/test_tar.py	
read	test/runtime/test_tensor.py	
read	test/runtime/test_tensor_cores.py	
read	test/runtime/test_tensor_data.py	
read	test/runtime/test_tensor_io.py	
read	test/runtime/test_tensor_variable.py	
read	test/runtime/test_to_numpy.py	
read	test/runtime/test_transcendental.py	
read	test/runtime/test_uops.py	

Event	Path or command	Hash
read	test/runtime/test_validate_with_cpu.py	
read	test/runtime/test_wait_loop.py	
read	test/runtime/test_winograd.py	
read	test/runtime/test_zero_copy.py	
read	test/speed/external_test_copy_speed.py	
read	test/speed/external_test_device_speed.py	
read	test/speed/external_test_specific_conv.py	
read	test/speed/external_test_speed_v_torch.py	
read	test/test_tiny.py	
read	test/testextra/__init__.py	
read	test/testextra/test_bench_log.py	
read	test/testextra/test_export_model.py	
read	test/testextra/test_f16_decompress.py	
read	test/testextra/test_fp8_linear.py	
read	test/testextra/test_hevc.py	
read	test/testextra/test_hk_fa.py	
read	test/testextra/test_lr_scheduler.py	
read	test/testextra/test_mockgpu.py	
read	test/testextra/test_tk.py	
read	test/web/test_webgpu.js	
read	tinygrad/codegen/__init__.py	
read	tinygrad/codegen/decomp/dtype.py	
read	tinygrad/codegen/decomp/op.py	
read	tinygrad/codegen/late/__init__.py	
read	tinygrad/codegen/late/coalesce.py	
read	tinygrad/codegen/late/gater.py	
read	tinygrad/codegen/opt/__init__.py	
read	tinygrad/codegen/opt/search.py	
read	tinygrad/device.py	
read	tinygrad/dtype.py	
read	tinygrad/engine/__init__.py	
read	tinygrad/engine/worker.py	
read	tinygrad/function.py	
read	tinygrad/helpers.py	
read	tinygrad/lm/__init__.py	
read	tinygrad/lm/__main__.py	
read	tinygrad/lm/cli.py	
read	tinygrad/lm/gguf.py	
read	tinygrad/lm/kernels/__init__.py	
read	tinygrad/lm/kernels/amd.py	
read	tinygrad/lm/model.py	
read	tinygrad/lm/serve.py	
read	tinygrad/mixin/dtype.py	
read	tinygrad/renderer/nir.py	
read	tinygrad/mixin/gradient.py	
read	tinygrad/mixin/movement.py	
read	tinygrad/mixin/rand.py	
read	tinygrad/nn/datasets.py	
read	tinygrad/nn/onnx.py	
read	tinygrad/nn/torch.py	

Event	Path or command	Hash
read	tinygrad/renderer/__init__.py	
read	tinygrad/renderer/amd/__init__.py	
read	tinygrad/renderer/amd/dsl.py	
read	tinygrad/renderer/amd/elf.py	
read	tinygrad/renderer/amd/generate.py	
read	tinygrad/renderer/amd/sqtt.py	
read	tinygrad/renderer/cstyle.py	
read	tinygrad/renderer/isa/x86.py	
read	tinygrad/renderer/wgsl.py	
read	tinygrad/runtime/__init__.py	
read	tinygrad/runtime/autogen/__init__.py	
read	tinygrad/runtime/autogen/am/fw.py	
read	tinygrad/runtime/autogen/am/navi_offsets.py	
read	tinygrad/runtime/autogen/am/pm4_nv.py	
read	tinygrad/runtime/autogen/am/pm4_soc15.py	
read	tinygrad/runtime/autogen/am/pmc.py	
read	tinygrad/runtime/autogen/am/regs.py	
read	tinygrad/runtime/autogen/am/sdma_4_0_0.py	
read	tinygrad/runtime/autogen/am/sdma_5_0_0.py	
read	tinygrad/runtime/autogen/am/sdma_6_0_0.py	
read	tinygrad/runtime/autogen/am/smu_13_0_0.py	
read	tinygrad/runtime/autogen/am/smu_13_0_12.py	
read	tinygrad/runtime/autogen/am/smu_13_0_6.py	
read	tinygrad/runtime/autogen/am/smu_14_0_2.py	
read	tinygrad/runtime/autogen/am/soc_11.py	
read	tinygrad/runtime/autogen/am/soc_12.py	
read	tinygrad/runtime/autogen/am/soc_9.py	
read	tinygrad/runtime/autogen/am/vega_offsets.py	
read	tinygrad/runtime/autogen/amd/__init__.py	
read	tinygrad/runtime/autogen/amd/cdna/__init__.py	
read	tinygrad/runtime/autogen/amd/cdna/enum.py	
read	tinygrad/runtime/autogen/amd/cdna/ins.py	
read	tinygrad/runtime/autogen/amd/cdna/operands.py	
read	tinygrad/runtime/autogen/amd/cdna/str_pcode.py	
read	tinygrad/runtime/autogen/amd/common.py	
read	tinygrad/runtime/autogen/amd/rdna3/__init__.py	
read	tinygrad/runtime/autogen/amd/rdna3/enum.py	
read	tinygrad/runtime/autogen/amd/rdna3/ins.py	
read	tinygrad/runtime/autogen/amd/rdna3/operands.py	
read	tinygrad/runtime/autogen/amd/rdna3/str_pcode.py	
read	tinygrad/runtime/autogen/amd/rdna4/__init__.py	
read	tinygrad/runtime/autogen/amd/rdna4/enum.py	
read	tinygrad/runtime/autogen/amd/rdna4/ins.py	
read	tinygrad/runtime/autogen/amd/rdna4/operands.py	
read	tinygrad/runtime/autogen/amd/rdna4/str_pcode.py	
read	tinygrad/runtime/autogen/amd_gpu.py	
read	tinygrad/runtime/autogen/amdgpu_kd.py	
read	tinygrad/runtime/autogen/avcodec.py	
read	tinygrad/runtime/autogen/bnxt.py	
read	tinygrad/runtime/autogen/comgr.py	

Event	Path or command	Hash
read	tinygrad/runtime/autogen/comgr_3.py	
read	tinygrad/runtime/autogen/corefoundation.py	
read	tinygrad/runtime/autogen/cuda.py	
read	tinygrad/runtime/autogen/ggml_common.py	
read	tinygrad/runtime/autogen/hip.py	
read	tinygrad/runtime/autogen/iokit.py	
read	tinygrad/runtime/autogen/kgsl.py	
read	tinygrad/runtime/autogen/libclang.py	
read	tinygrad/runtime/autogen/llvm.py	
read	tinygrad/runtime/autogen/llvm_qcom.py	
read	tinygrad/runtime/autogen/mesa.py	
read	tinygrad/runtime/autogen/metal.py	
read	tinygrad/runtime/autogen/mlx5.py	
read	tinygrad/runtime/autogen/nv.py	
read	tinygrad/runtime/autogen/io_uring.py	
read	tinygrad/runtime/autogen/nv_570.py	
read	tinygrad/runtime/autogen/nv_580.py	
read	tinygrad/runtime/autogen/nv_regs/__init__.py	
read	tinygrad/runtime/autogen/nv_regs/dev_bus.py	
read	tinygrad/runtime/autogen/nv_regs/dev_falcon_second_pri.py	
read	tinygrad/runtime/autogen/nv_regs/dev_falcon_v4.py	
read	tinygrad/runtime/autogen/nv_regs/dev_fb.py	
read	tinygrad/runtime/autogen/nv_regs/dev_fbif_v4.py	
read	tinygrad/runtime/autogen/nv_regs/dev_fsp_pri.py	
read	tinygrad/runtime/autogen/nv_regs/dev_gc6_island.py	
read	tinygrad/runtime/autogen/nv_regs/dev_gsp.py	
read	tinygrad/runtime/autogen/nv_regs/dev_mmu.py	
read	tinygrad/runtime/autogen/nv_regs/dev_riscv_pri.py	
read	tinygrad/runtime/autogen/nv_regs/dev_sec_pri.py	
read	tinygrad/runtime/autogen/nv_regs/dev_therm.py	
read	tinygrad/runtime/autogen/nv_regs/dev_vm.py	
read	tinygrad/runtime/autogen/nv_regs/nv_ref.py	
read	tinygrad/runtime/autogen/nvjitlink.py	
read	tinygrad/runtime/autogen/nvrtc.py	
read	tinygrad/runtime/autogen/openssl.py	
read	tinygrad/runtime/autogen/qcom_dsp.py	
read	tinygrad/runtime/autogen/rocprow.py	
read	tinygrad/runtime/autogen/sqtt.py	
read	tinygrad/runtime/autogen/webgpu.py	
read	tinygrad/runtime/ops_amd.py	
read	tinygrad/runtime/ops_cl.py	
read	tinygrad/runtime/ops_cuda.py	
read	tinygrad/runtime/ops_disk.py	
read	tinygrad/runtime/ops_dsp.py	
read	tinygrad/runtime/ops_hip.py	
read	tinygrad/runtime/ops_metal.py	
read	tinygrad/runtime/ops_npy.py	
read	tinygrad/runtime/ops_null.py	
read	tinygrad/runtime/ops_nv.py	
read	tinygrad/runtime/ops_qcom.py	

[illegible]

Event	Path or command	Hash
process	/var/tmp/kv-repro/010-tinygrad-tinygrad/src/./env/bin/python -c from multipr...	
process	/var/tmp/kv-repro/010-tinygrad-tinygrad/src/./env/bin/python -c from multipr...	
process	/var/tmp/kv-repro/010-tinygrad-tinygrad/src/./env/bin/python -c from multipr...	
process	/var/tmp/kv-repro/010-tinygrad-tinygrad/src/./env/bin/python -c from multipr...	
process	/var/tmp/kv-repro/010-tinygrad-tinygrad/src/./env/bin/python -c from multipr...	
process	/var/tmp/kv-repro/010-tinygrad-tinygrad/src/./env/bin/python -c from multipr...	
process	/var/tmp/kv-repro/010-tinygrad-tinygrad/src/./env/bin/python -c from multipr...	
process	/var/tmp/kv-repro/010-tinygrad-tinygrad/src/./env/bin/python -c from multipr...	
process	/var/tmp/kv-repro/010-tinygrad-tinygrad/src/./env/bin/python -c from multipr...	
process	/var/tmp/kv-repro/010-tinygrad-tinygrad/src/./env/bin/python -c from multipr...	
process	/var/tmp/kv-repro/010-tinygrad-tinygrad/src/./env/bin/python -c from multipr...	
process	/var/tmp/kv-repro/010-tinygrad-tinygrad/src/./env/bin/python -c from multipr...	
process	/var/tmp/kv-repro/010-tinygrad-tinygrad/src/./env/bin/python -c from multipr...	
process	/var/tmp/kv-repro/010-tinygrad-tinygrad/src/./env/bin/python -c from multipr...	
process	/var/tmp/kv-repro/010-tinygrad-tinygrad/src/./env/bin/python -c from multipr...	
process	/var/tmp/kv-repro/010-tinygrad-tinygrad/src/./env/bin/python -c from multipr...	
process	/var/tmp/kv-repro/010-tinygrad-tinygrad/src/./env/bin/python -c from multipr...	
process	/var/tmp/kv-repro/010-tinygrad-tinygrad/src/./env/bin/python -c from multipr...	
process	/var/tmp/kv-repro/010-tinygrad-tinygrad/src/./env/bin/python -c from multipr...	
process	clang -c -x c -O2 -fPIC -ffreestanding -fno-math-errno -nostdlib -fno-ident --t...	
process	./env/bin/python -u examples/beautiful_mnist.py	
process	awk { print; fflush() } /test_accuracy: +[0-9.]+%/ { match(\$0, /test_accura...	
process	tr \r \n	
process	/var/tmp/kv-repro/010-tinygrad-tinygrad/src/./env/bin/python -c from multipr...	
process	/var/tmp/kv-repro/010-tinygrad-tinygrad/src/./env/bin/python -c from multipr...	
process	/var/tmp/kv-repro/010-tinygrad-tinygrad/src/./env/bin/python -c from multipr...	
process	/var/tmp/kv-repro/010-tinygrad-tinygrad/src/./env/bin/python -c from multipr...	
process	/var/tmp/kv-repro/010-tinygrad-tinygrad/src/./env/bin/python -c from multipr...	
process	/var/tmp/kv-repro/010-tinygrad-tinygrad/src/./env/bin/python -c from multipr...	
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process	/var/tmp/kv-repro/010-tinygrad-tinygrad/src/./env/bin/python -c from multipr...	
process	/var/tmp/kv-repro/010-tinygrad-tinygrad/src/./env/bin/python -c from multipr...	
process	/var/tmp/kv-repro/010-tinygrad-tinygrad/src/./env/bin/python -c from multipr...	
process	/var/tmp/kv-repro/010-tinygrad-tinygrad/src/./env/bin/python -c from multipr...	
process	/var/tmp/kv-repro/010-tinygrad-tinygrad/src/./env/bin/python -c from multipr...	
process	/var/tmp/kv-repro/010-tinygrad-tinygrad/src/./env/bin/python -c from multipr...	
process	/var/tmp/kv-repro/010-tinygrad-tinygrad/src/./env/bin/python -c from multipr...	
process	/var/tmp/kv-repro/010-tinygrad-tinygrad/src/./env/bin/python -c from multipr...	
process	/var/tmp/kv-repro/010-tinygrad-tinygrad/src/./env/bin/python -c from multipr...	

C Run history

Invocation 1, OK, 5 minutes 58 seconds, 1113 output lines, started 2026-10-07T07:37:16

metric_hash 8a637d2e9bc282b654d8269d02428f7cceb4b7e11d69c69893eb90bcc45cb75e
run_digest 07725a52a9b255f999240a81e10d5ef0315a937aafeac32365b8e540e51f16c6
anchored at run end

D Raw seal fields and hashes

Seal

```
data_hash      7a4a9afa73e75c11dd8814c9f98bd317f848d771a8272d361e229ef70aa583e7
signed_message_hash 44ad70a27d9c0615444b76a11a0bcd4c1b3c59ae320e7823bb1cf6d417b58669
nonce         2afc9e6e867daf9d1dd12eca2cf6a354
signed_at     2026-10-07T07:43:17.469358+00:00
sealed_at     2026-10-07T07:43:17.528427541Z
checkout_bundle bae3e78018887e89e095c6f54251e33fe7e98e443178b2ea040bb18be22af6a1
```

Signature (base64)

```
bcWnAjp8kT9z6tmpom1CcS0+KPPaVtJCD5X7xrhbkPwgN9GeDiMZYWE1/CoAMC06+paK0l2+ISGwS0J8HFdLeBHx
yQaWP/0/135mTr1J25/bJoMzWceYsbdT0SK5UbUHZfEvPA3QIgFrB45j/ESurGR15wBRS0vnzkebj+dcvC4wE8Ge
sZskYvybaJNlEI/LwkSc7M8YM1r7sYhnkLaqC5XrUr4Bz5Uyjf0kuOKpR4BJz5pewswaNnLzR6ZidPiIPSAjy7Bz
RdF8ck0l5Bv3mHRR5c9AnWlKvT5cpaQl89l5w9I2uQh2+w8FjNaotc1hY0jXnaZSp6r2VE0m8FXHjdpBarB6w16d
SL955YLDjTQscucqASpMB1nxA/F/xNDzZtwdvdxPgqHm0wBpDF5CFRbedxNtTA2dxS1DJVXV7c4EVWJjmRBpkScj
QFRfAcxI9ZZ68md5Sw5vyWxZsr0Clza85wbV3qu4g1El0BV2smkuX8dxkEqmxdMcX3148Qu/n6RRJjwWNJBQzey
3nbW4GnRhnQZw5q/NBG08PXkPfgeHmM6z8eC5TSXVyvej1w0PzXkhLp0pd8tWMV6kTfUzdhsbH0QI4Xwz36FGC1r
kZH0lLAXQcTdxUSJUCwkcI96o5skBjtLiLXfpqj4HUBLSnIY6PLW6GrJlhaQ1X+05RY=
```

Public key

```
-----BEGIN PUBLIC KEY-----
MIICIjANBgkqhkiG9w0BAQEFAAOCAg8AMIICCGKCAgEA1QL3YyMgeShx8IhTC7jb
sM0w2BnRXgsBn3+SMXKjtwHATqdekmlPlcjayJxD5fcProPPCqp/LouFz5CUZzRi
6+8FTv70dHuIz5A4LAT4vcgwP+iWscPc3HzpmjpsKbTm0emqgp6RB6GNNpXyA5X
7qU95EFEisX/GfoS9ToxtqPBwCcVWSPBX8efec4TC58m96Fsp4mC8KICeadbIbHN
av0JjdTJ//CM0WZSIbdzE6qvEbNfcqDB5tNU7oHfYvBeNLrvPU9TAqdbtozdHq+D
AhfTRuJNgWEkjwD5lboVhi4qlkweSOEMTKT3ye9Il5xEaJn4bku4eKqeS7VSzt3o
XbUcmr2H5mKcKsvt6bXmX6ajAmgi900Cm9QJlaQkJM6z2iITnyvDMgueJtFxQpC9
HNBPGPZ242NH8JgE6CNF7ToL9syasS67u7UMgIcNgeNHnCy9bLgPU5annzP9gN2l
6HEMldkXNdKQRb1rDwyEJf3N/50YHnXXPmuf5sb8R/pxWNqtF04R5UGVV92qd2Mi
1GN7lE3mbQRbPsLL/JApDXMNzNikaFfBsQK7iKxr7vIaTeZckFZWfThcYNT7Hasg
+eE4bf/olCHBCf1VIDCKDxoehdZ0WRRcgM9iK3I/0zZdaBsgdc8WuJRmb8LPP9v
8Jap/1PwjNcNku5cdW3gHLcCAwEAAQ==
-----END PUBLIC KEY-----
```

Run 1 outputs

```
stdout 7665aaab4e29a2aa052f20e24a8a4f77a2377079b862cfb137bd1611c3970657
stderr e3b0c44298fc1c149afb4c8996fb92427ae41e4649b934ca495991b7852b855
metrics 8a637d2e9bc282b654d8269d02428f7cceb4b7e11d69c69893eb90bcc45cb75e
digest 07725a52a9b255f999240a81e10d5ef0315a937aafec32365b8e540e51f16c6
kv_run.sh (unchanged)
pre 2758e356ea5975183aabfdcf2cd02df88b990cb70423fe3df75371cd4f2a7f84
post 2758e356ea5975183aabfdcf2cd02df88b990cb70423fe3df75371cd4f2a7f84
```

E BibTeX for [1]

```
@misc{keita2026computerscienceconferencesrequire,
  title={Computer Science Conferences Should Require Nonrepudiable Experimental Results},
  author={Mamadou K. Keita and Christopher Homan},
  year={2026},
  eprint={2605.08586},
  archivePrefix={arXiv},
  primaryClass={cs.CR},
  url={https://arxiv.org/abs/2605.08586},
}
```